

Analysis Report

Plant Name	Essens Renewable Pvt Ltd		
Installer Company	AMPLUS IRU PRIVATE LIMITED		
Site In charge	Amit Jadhav	Phone No	8657894046
Plant Address	Amplus IRU Pvt Ltd, Thoothkudi District, Pin code-628001, Tamil Nadu		
Inverter SN	I23B1700303		
SG-Service Engineer	Mr. Ganesh.	Visited Date	NA

Subject:

The inverter model SG3125HV-32 was tripping due to AC overvoltage.

Description:

1. As reported by the site team, Block-3 Inverter-1 was tripping frequently on AC overvoltage during the morning hours. Based on information, the SG technical engineer collected the inverter data along with ACB trip related details.
2. Based on the analysis of the available inverter fault records, it was observed that the inverter tripped multiple times at different intervals due to abnormal frequency, AC overvoltage, islanding protection, and other related events. The corresponding fault records are presented in Figure.

Devices	Event name	Event Lev	Generation time
PV INVERTER_1	AC overvoltage of unit 1	Fault	07-09-2026 07:45
PV INVERTER_1	AC overvoltage of unit 1	Fault	06-09-2026 12:27
PV INVERTER_1	Abnormal frequency of unit 1	Fault	06-09-2026 12:27
PV INVERTER_1	AC overvoltage of unit 2	Fault	06-09-2026 12:27
PV INVERTER_1	Abnormal frequency of unit 2	Fault	06-09-2026 12:27
PV INVERTER_1	AC overvoltage of unit 1	Fault	06-09-2026 07:49
PV INVERTER_1	AC overvoltage of unit 1	Fault	04-09-2026 07:42
PV INVERTER_1	AC undervoltage of unit 1	Fault	03-09-2026 18:01
PV INVERTER_1	Abnormal frequency of unit 1	Fault	03-09-2026 18:01
PV INVERTER_1	AC overvoltage of unit 2	Fault	02-09-2026 07:42
PV INVERTER_1	Abnormal frequency of unit 2	Fault	02-09-2026 07:42
PV INVERTER_1	AC overvoltage of unit 1	Fault	02-09-2026 07:41
PV INVERTER_1	Abnormal frequency of unit 1	Fault	02-09-2026 07:41

Figure: Inverter Fault Records

3. Analysis of the inverter FLR data corresponding to the fault-triggered timestamp indicates a sudden abnormal reduction in grid frequency, with the frequency reaching approximately 46.1 Hz immediately before the trip event. The recorded frequency trend is presented in Figure (a).

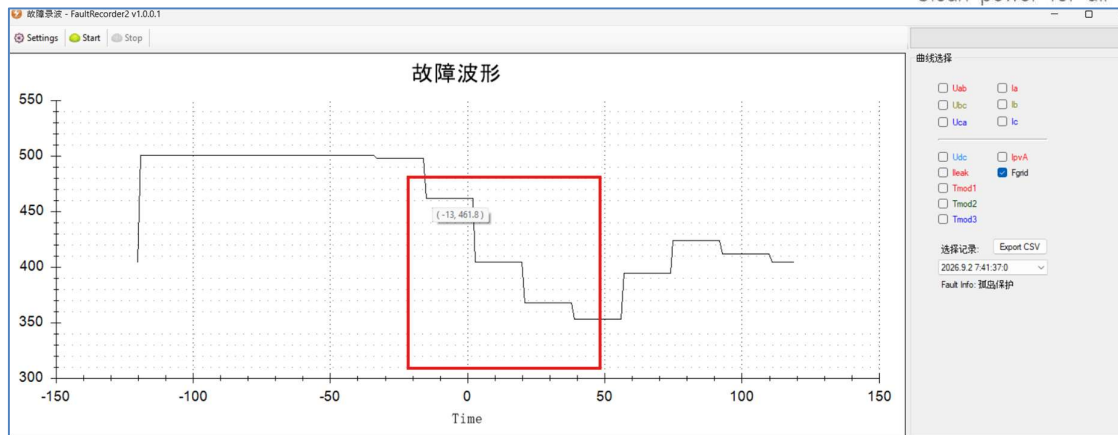


Fig: Graph Record of frequency

4. Furthermore, the AC voltage exhibited a significant rise, reaching approximately 978.5 V. As a result of the elevated AC voltage, the inverter operated its AC Overvoltage protection and subsequently tripped.

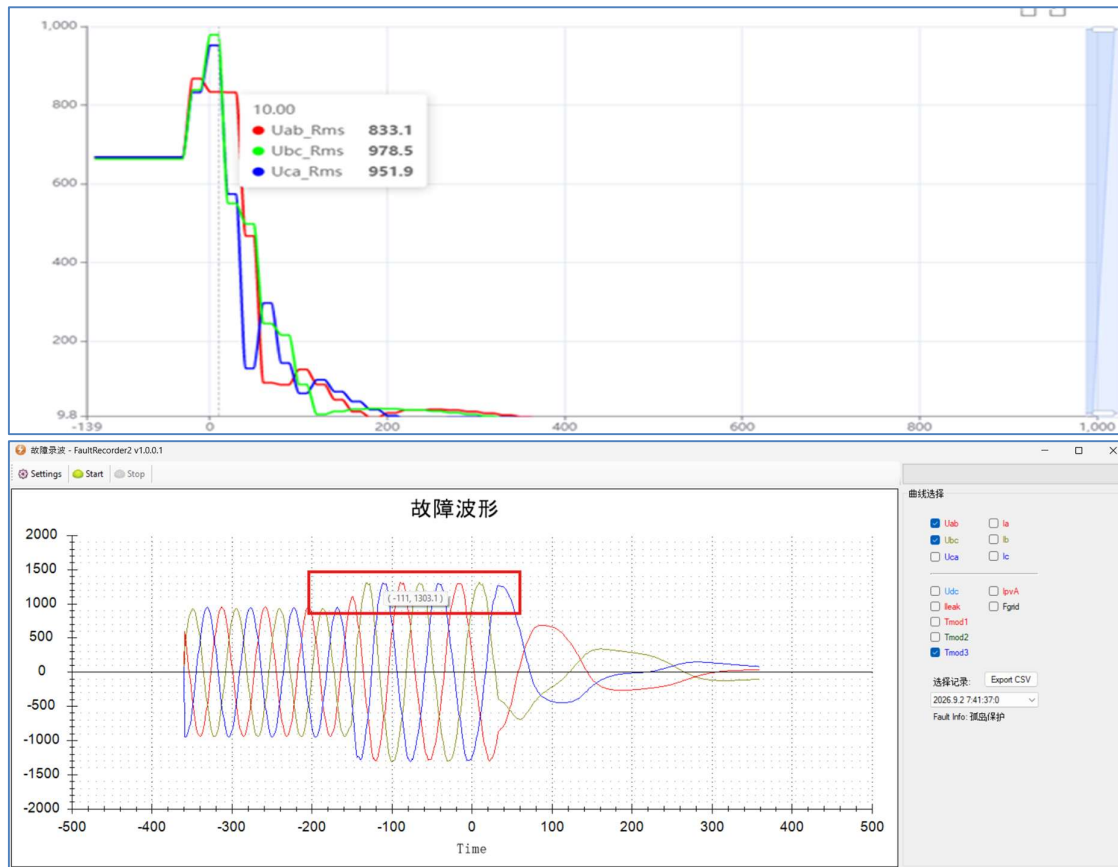


Fig: Graph Record of AC voltage

5. During the recorded event, the AC line-to-line RMS voltage exceeded the permissible voltage of 660 V. The maximum recorded voltage was approximately 978.5 V on Ubc, while Uca reached approximately 951.9 V and Uab reached approximately 833.1 V.
6. It has been observed that the ACB breaker was tripped before the inverter tripping with Ac overvoltage. And observed the ACB tripped by indicating the “Ig - fault”. Refer to the image

below.



Fig: ACB tripped image

7. Based on the analysis of the ACB and inverter event records, the ACB was observed to trip initially on Ig (ground-fault) protection, indicating the detection of abnormal residual/ground-fault current in the AC circuit. Subsequently, the inverter recorded abnormal grid conditions, including a frequency drop to approximately 46.1 Hz and a transient AC overvoltage, with the maximum line-to-line voltage reaching approximately 978.5 V. The inverter then tripped on abnormal-frequency and AC-overvoltage protection.
8. The event sequence indicates that the ground-fault condition and subsequent ACB Ig operation preceded the inverter protection trips. Therefore, the inverter abnormal-frequency and AC-overvoltage faults are considered consequential responses to the AC-side disturbance following the ACB operation, rather than the initiating cause of the event.
9. Accordingly, the primary cause of the incident is considered to be an abnormal AC-side ground-fault condition that resulted in ACB Ig protection operation and subsequent disturbance of the inverter AC supply. Further investigation of the AC cables, cable terminations, transformers, switchgear and grounding system is recommended to identify the exact source of the ground fault.

Conclusion:

Based on the analysis of inverter data and ACB fault records, along with the recorded voltage trend, the event was initiated by an AC-side abnormal-fault condition that resulted in the ACB operating on Ig (ground-fault) protection. And the inverter experienced an abnormal AC-side disturbance, including a significant frequency drop to approximately 46.1 Hz and a transient overvoltage, with the maximum line-to-line voltage reaching approximately 978.5 V against the permissible 660 V level. The inverter subsequently tripped on abnormal-frequency and AC-overvoltage protection.

Further verification of the AC cables, terminations, transformers, grounding system, ACB operation and upstream grid/measurement records is recommended to identify the exact source of the initial ground-fault condition and associated voltage disturbance.

Sungrow Service Engineer/Ganesh K	Audited By Zonal Manager/ Venkatesh. P
Date: 09-09-2026	Date: 09-09-2026